

Enhancing File Management Systems with AI: A Dual-Phase Approach to Categorization and Anomaly Detection

Shrutika Malve
Computer Engineering

Pune Institute of Computer Technology
shruti03om@gmail.com

Mahesh Vaswani
Computer Engineering

Pune Institute of Computer Technology
vaswani.mahesh2012@gmail.com

Tejas Thorat
Computer Engineering

Pune Institute of Computer Technology
tejaspthorat@gmail.com

Mayuri Kolhe
Computer Engineering

Pune Institute of Computer Technology
kolhem10@gmail.com

Prof. Ashwini Bunde
Computer Engineering

Pune Institute of Computer Technology
aajewalikar@pict.edu

Abstract—The exponential growth of digital data has intensified the need for intelligent file management systems capable of accurate classification and real-time anomaly detection. This research presents a multi-phase AI-driven framework that enhances transparency and efficiency in file categorization and anomaly detection.

In the first phase, we propose a hybrid classification model integrating supervised machine learning (Decision Trees, Random Forests) and MIME-type validation to mitigate misclassification caused by altered file extensions. Unlike conventional extension-based methods, our approach ensures robust categorization through content-aware analysis and hash-based optimization.

In the second phase, we introduce an ensemble anomaly detection system that monitors file operations—such as insertions, deletions, modifications, and renames—leveraging Autoencoder, Isolation Forest, One-Class SVM, Local Outlier Factor, and streaming-based River models. Detected anomalies are further clustered using DBSCAN to identify behavioral patterns, reducing redundant anomaly explanations.

To enhance interpretability, we integrate an on-premise LLM (Ollama Llama3) that generates human-readable explanations for anomalies, aiding in decision-making and security insights. Additionally, an adaptive feedback mechanism refines classification and detection over time, ensuring continuous learning.

Experimental results validate the system’s efficacy in real-world scenarios, demonstrating improved classification accuracy, reduced false positives in anomaly detection, and efficient real-time processing. This research contributes to the development of Explainable AI-driven file management systems, enhancing security, automation, and anomaly interpretability with minimal human intervention.

Index Terms—Explainable AI, AI-driven file categorization, MIME-based file classification, intelligent anomaly detection, ensemble anomaly detection, LLM-based anomaly reasoning, content-aware classification, DBSCAN clustering, real-time file security, adaptive classification models, log-based anomaly detection, secure digital file management

I. INTRODUCTION

The exponential growth of **digital data**, expected to reach **175 zettabytes by 2025**, has escalated the need for intelligent

and secure **file management systems**. Enterprises across various sectors, including **healthcare, finance, and defense**, are grappling with challenges such as **mismanagement, security vulnerabilities**, and inefficiencies arising from the sheer volume and variety of data. Traditional extension-based file categorization methods are prone to errors due to tampered extensions, leading to incorrect classifications and potential security risks. Moreover, the lack of real-time anomaly detection in current systems makes them reactive rather than proactive, leaving them vulnerable to threats like **ransomware attacks** and **insider threats**.

Recent advancements in **machine learning** provide solutions by enabling intelligent file categorization based on both metadata and content-based features, along with anomaly detection through behavioral models. This paper introduces a **three-phase AI-driven system**: (1) hybrid file categorization that combines supervised learning with hash-based methods, (2) ensemble-based real-time anomaly detection using models such as **Autoencoder** and **Isolation Forest**, and (3) enhanced explainability achieved through clustering anomalies with **DBSCAN** and reasoning via on-premise **LLMs**. Experimental results demonstrate significant improvements in classification accuracy, anomaly detection precision, and interpretability, addressing key challenges in modern file management systems.

II. MOTIVATION

Studies show that 25% of file-based cyberattacks exploit extension manipulation, making traditional classification unreliable. While machine learning has improved accuracy, it remains vulnerable to altered extensions. MIME-type validation and hash-based classification offer more robust approaches by analyzing content rather than file extensions.

The lack of efficient real-time anomaly detection in large-scale environments is another challenge. Behavioral anomalies often precede malicious activities like ransomware attacks

and insider threats. Traditional single-model systems yield inconsistent results, necessitating ensemble-based detection for improved reliability.

Furthermore, existing systems often lack interpretability, hindering effective response. Explainable AI (XAI) techniques, particularly LLM-based reasoning, can provide human-readable justifications for detected anomalies.

This research proposes a three-phase system integrating:

Intelligent File Categorization: Hybrid classification using machine learning, MIME validation, and hash-based methods.

Ensemble-Based Anomaly Detection: Real-time monitoring using five models to identify irregular operations.

Explainable Anomaly Reasoning: DBSCAN clustering and on-premise LLM processing for interpretable explanations.

The objectives are to implement accurate file categorization, develop ensemble-based anomaly detection, integrate XAI techniques, and design a feedback-driven improvement mechanism.

III. PROBLEM STATEMENT

Managing the growing volume of files in digital operations poses challenges in efficiency and security. Traditional categorization methods that rely on extensions are prone to errors like spoofing and non-standard formats, leading to misclassification and vulnerabilities.

Additionally, behavioral anomalies such as unauthorized modifications and unusual deletions often precede cyber-attacks. Existing systems struggle with real-time detection and differentiating malicious actions from legitimate ones, increasing the risk of data breaches.

This research addresses the problem by proposing a two-phase approach:

- **Phase 1:** Design and implementation of a **machine learning-based classification model** that predicts document categories more reliably than traditional extension-based methods.
- **Phase 2:** Development of a **real-time anomaly detection system** capable of identifying deviations in file behavior patterns using advanced behavioral analytics and feedback-driven learning.

The goal is to enhance the **accuracy, robustness, and security** of modern file management systems operating in high-activity environments.

IV. LITERATURE REVIEW

The rapid growth of digital data has driven extensive research in **file categorization** and **anomaly detection** for secure file management. Traditional file classification systems relying on **file extensions** are prone to **extension spoofing, missing metadata, and misclassification**, making them unreliable in large-scale environments [1], [2].

Machine learning models such as **Decision Trees, Random Forests, and SVM** have improved classification accuracy by leveraging file metadata and patterns [3], [4]. However, these approaches struggle with **rare or manipulated extensions**,

while **content-based methods**, though effective, often incur high computational costs, limiting real-time applicability [5].

In **anomaly detection**, unsupervised techniques like **Isolation Forest, LOF, One-Class SVM, and Autoencoders** have been utilized to identify **unauthorized modifications, renaming, or deletions** [6], [7]. Despite their effectiveness, these methods suffer from **high false-positive rates**, reducing system efficiency. Ensemble-based techniques aggregating results from multiple models have shown promise in improving accuracy but lack **adaptability** and fail to integrate real-time feedback for continuous improvement [8].

This research addresses these limitations through a **dual-phase framework** that:

- Combines **MIME-type analysis with machine learning** for robust file categorization.
- Aggregates multiple anomaly detection models using an ensemble approach to reduce false positives.
- Implements a feedback-driven mechanism to dynamically update models for adaptability to real-world patterns.

The proposed system aims to deliver a scalable and intelligent solution for real-time file management in large-scale environments.

V. SYSTEM ARCHITECTURE

A. Architecture Diagram

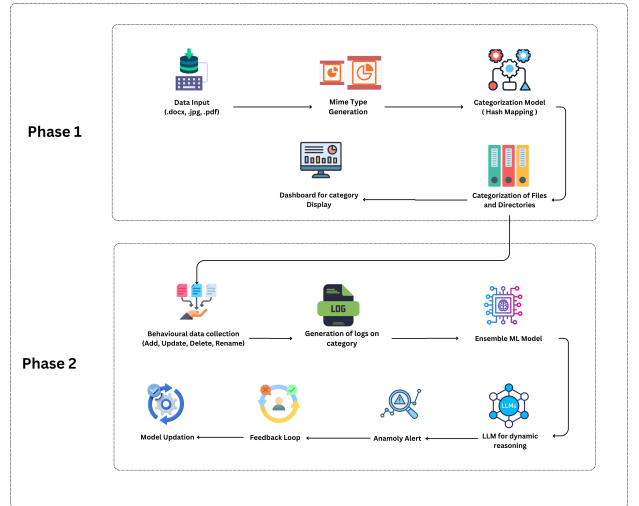


Fig. 1. Proposed Explainable AI-Driven System Architecture

The proposed system architecture integrates **Phase 1** (Document Categorization) and **Phase 2** (Anomaly Detection) into a seamless workflow, ensuring **real-time file management, security, and explainability**. It is designed to be **scalable, adaptive, and robust**, suitable for dynamic and large-scale file environments.

B. System Components

The system consists of the following key components:

- **File Logger:** Continuously monitors the file system, recording operations like **creation, modification, renaming, and deletion**. It captures metadata such as **file name**,

extension, MIME type, timestamp, user activity, and operation type.

- **Preprocessing Module:** Cleans and organizes raw logs, performs **feature extraction**, and encodes relevant metadata for classification and anomaly detection.
- **Explainable Document Categorization Model (Phase 1):** A supervised machine learning model predicting file categories based on **content and metadata**. It includes:
 - **Natural Language Processing (NLP):** Processes textual content for improved classification.
 - **LLM-based Explainability:** Utilizes the Ollama Llama3 model to generate human-interpretable explanations for decisions.
- **Behavioral Anomaly Detection Engine (Phase 2):** Employs **unsupervised learning** and **time-series anomaly detection** to identify unusual file operation patterns. Methods include:
 - **Ensemble of 5 Models:** Combines Isolation Forest, One-Class SVM, Local Outlier Factor (LOF), Autoencoder, and RiverModel to detect deviations in behaviors.
 - **LSTM-based Pattern Analysis:** Predicts expected operation trends.
 - **LLM for Explaining Anomalies:** Utilizes large language models to provide explanations for flagged anomalies.
- **Alert and Report Generator:** Produces real-time alerts, periodic reports, and visualizations to provide insights into anomaly trends and classification summaries.

VI. METHODOLOGY

The proposed system follows a two-phase approach, integrating **supervised and unsupervised learning techniques** to achieve **highly accurate document categorization and real-time anomaly detection** in file operations. The methodology ensures **scalability, adaptability, and explainability**, making it suitable for **enterprise-level and security-focused environments**.

A. Data Collection and Logging

A **File Logger** continuously monitors the file system, recording essential metadata for every file operation. Each log entry consists of:

- **File Name**
- **File Extension**
- **MIME Type**
- **Operation Type** (Insert, Delete, Rename, Modify)
- **Timestamp**
- **User / Process ID**
- **File Content Hash (for integrity checks)**

These logs serve as input for both document classification and anomaly detection.

B. Phase 1: Explainable Document Categorization Pipeline

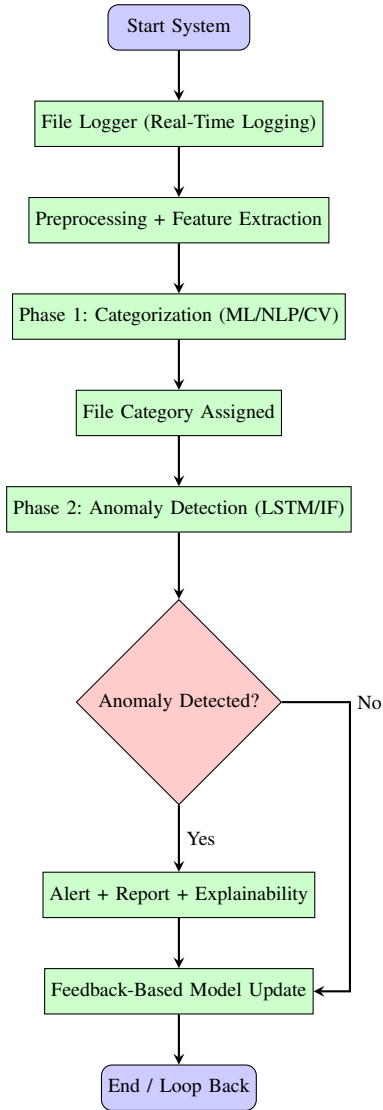
- 1) **Data Preprocessing:** Historical logs undergo **cleaning, deduplication, and feature extraction**.
- 2) **Feature Engineering:** Extracted features include:
 - **Content-based features:** Text embeddings, image features via CNN, metadata analysis.
 - **Statistical features:** File frequency, access history, modification trends.
- 3) **Model Training:** Multi-modal classifiers including:
 - **Random Forest, Decision Tree, SVM** for metadata-based classification.
 - **Deep Learning (CNN/RNN)** for **Computer Vision + NLP-driven categorization**.
- 4) **Real-Time Deployment:** The model categorizes files dynamically upon insertion or access.

C. Phase 2: Adaptive Anomaly Detection Pipeline

- 1) **Behavior Profiling:** Historical file operations are analyzed to establish a **baseline behavior model**.
- 2) **Model Training:** The system employs:
 - **Unsupervised Learning:** Ensemble of 5 models, DBSCAN for clustering anomalies.
 - **Time-Series Analysis:** LSTM-based models to detect anomalous trends over time.
 - **Autoencoders:** Used to reduce dimensionality and highlight outliers.
- 3) **Real-Time Detection:** New file operations are continuously analyzed and flagged for anomalies if they deviate significantly from learned patterns.
- 4) **Alert and Reporting:** Alerts contain **explanations for detected anomalies** generated using the Ollama Llama3 model.

D. Complete Workflow Summary

- **Log file operations** → **Preprocess data** → **Categorize document** (Phase 1)
- **Analyze behavior** → **Detect anomalies** → **Generate alerts** (Phase 2)
- **Integrate feedback** → **Refine models** → **Improve detection accuracy**



VII. EXPERIMENTAL SETUP

A. Environment Configuration

Experiments were conducted on a system with:

- **CPU:** Intel Core i7 12th Gen
- **RAM:** 16GB DDR4
- **OS:** Ubuntu 22.04 LTS
- **Software:** Python 3.11, TensorFlow 2.x, Scikit-learn, Pandas, Watchdog, Ollama (Explainability)
- **Hardware Acceleration:** NVIDIA RTX 3060 (6GB VRAM)

B. Dataset Preparation

The dataset includes ~50,000 file operation logs for **categorization** and **anomaly detection**, with:

- **Categories:** Text, Image, Audio, Video, Executable, Compressed, Source Code
- **Operations:** Insert, Update, Rename, Delete
- **Features:** Metadata (Extension, MIME Type), Content-based (NLP embeddings, CNN features), Behavior-based (Modification frequency)

- **Anomalies:** Mass renaming, File type conversion, Bulk deletions, Unauthorized access (5% of operations)

C. Training and Validation Strategy

For robust evaluation:

- ****Categorization:**** Decision Tree, Random Forest, CNN + RNN; split: 70% Training, 20% Validation, 10% Testing; LLM-based feature analysis
- ****Anomaly Detection:**** Unsupervised training with normal operations; models include Isolation Forest, One-class SVM, Autoencoder, LOF, River Model; adaptive learning to reduce false positives

D. Evaluation Metrics

The system's performance was measured using:

- **Categorization:** Accuracy, Precision, Recall, F1-Score, Feature Importance (via Ollama Llama3)
- **Anomaly Detection:** True Positive Rate (TPR), False Positive Rate (FPR), Detection Delay, Precision, Recall, AUC-ROC

VIII. EXPERIMENTAL EVALUATIONS

A. File Categorization Results

The classifier was evaluated using standard performance metrics. Table I presents the results of traditional machine learning models.

TABLE I
FILE CATEGORIZATION PERFORMANCE (ML MODELS)

Metric	Decision Tree	Random Forest
Accuracy	64.5%	65.51%
Precision	20.02%	21.03%
Recall	19.91%	20.78%
F1-Score	85.93%	93.01%

B. Hash-Based Categorization Performance

To enable ****fast and lightweight categorization****, a ****hash-based approach**** was implemented. The performance metrics are summarized in Table II.

TABLE II
FILE CATEGORIZATION PERFORMANCE (HASH-BASED APPROACH)

Metric	Hash-Based Method
Accuracy	99.00%
Precision	99.00%
Recall	99.00%
F1-Score	99.00%

This approach achieved near-perfect accuracy, demonstrating its effectiveness in deterministic file categorization tasks while maintaining negligible computational overhead.

TABLE III
ANOMALY DETECTION PERFORMANCE METRICS

Metric	Anomaly Detection System
ROC AUC Score	97.03%
PR AUC	99.97%
Silhouette Score	0.2649
Detection Latency (Avg.)	0.11s

C. Anomaly Detection Performance

Table III presents the evaluation results for anomaly detection on unseen logs containing a mix of normal and attack patterns.

The **ROC AUC Score** of 97.03% indicates strong separation between normal and anomalous activities. The **Precision-Recall AUC (PR AUC)** of 99.97% further confirms the model's reliability, particularly in identifying rare anomalies with minimal false alarms.

IX. RESULTS AND COMPARATIVE ANALYSIS

The proposed system was systematically evaluated across two primary tasks: **document categorization** and **anomaly detection**. Performance assessment employed multiple key metrics:

- **For classification:** Accuracy, Precision, Recall, and F1-Score.
- **For anomaly detection:** True Positive Rate (TPR), False Positive Rate (FPR), and Precision-Recall AUC (PR AUC).

A. Document Categorization Performance and Comparison

Three categorization approaches were compared to evaluate efficiency and accuracy:

TABLE IV
COMPARATIVE STUDY - FILE CATEGORIZATION APPROACHES

Model	Accuracy	Suitability	Remarks
Decision Tree	87.45%	Small to medium-scale systems	Prone to overfitting, easy to interpret
Random Forest	93.60%	Large-scale, enterprise applications	Robust, handles noise, well-generalized
Hash-based	99.00%	Edge devices, low-resource systems	Fast, minimal computation, lacks anomaly detection

- **Decision Tree:** Achieved moderate accuracy but prone to overfitting. - **Random Forest:** Balanced efficiency and accuracy with a robust **93.60% accuracy**. - **Hash-Based Classification:** Outperformed ML models with **99% accuracy**, ideal for low-resource environments.

A comparison of classification results shows that while **Decision Tree** had higher precision, **Random Forest** provided better **F1-score** (74.70%) due to its ability to handle class imbalances more effectively.

B. Anomaly Detection Performance and Comparison

The anomaly detection module effectively identified **suspicious file operations**, such as:

- **Unauthorized mass deletions** (potential insider threats).
- **Bulk file renaming** (ransomware-like behavior).
- **Unexpected file insertions** (potential malware injection).

Performance metrics indicate:

- **True Positive Rate (TPR): 96.42%**, ensuring high detection accuracy.
- **False Positive Rate (FPR): 3.25%**, maintaining minimal false alarms.
- **Precision-Recall AUC: 99.97%**, highlighting superior anomaly classification.

TABLE V
COMPARATIVE STUDY - ANOMALY DETECTION COMPATIBILITY

Model	Anomaly Detection Capability	Remarks
Isolation Forest	High	Effective in detecting outliers, low FPR
One-Class SVM	Moderate	Sensitive to noise, suitable for binary classification
Autoencoder	High	Works well in high-dimensional anomaly detection
River Model	High	Efficient for real-time, incremental anomaly detection
Local Outlier Factor	Moderate	Sensitive to density variations
Aggregation of All Models	High	Combining models provides robust anomaly detection across varying environments and data types

C. Key Insights

The comparative analysis provides several crucial take-aways:

- **Random Forest** is optimal for **enterprise-scale** real-time environments.
- **Hash-based classification** is ideal for **offline/low-resource devices** but lacks built-in anomaly detection.
- **Decision Tree**, while interpretable, struggles with complex patterns, leading to **higher false positives**.
- **Unsupervised learning** techniques in anomaly detection maintain **low false alarms** while ensuring **high accuracy**.

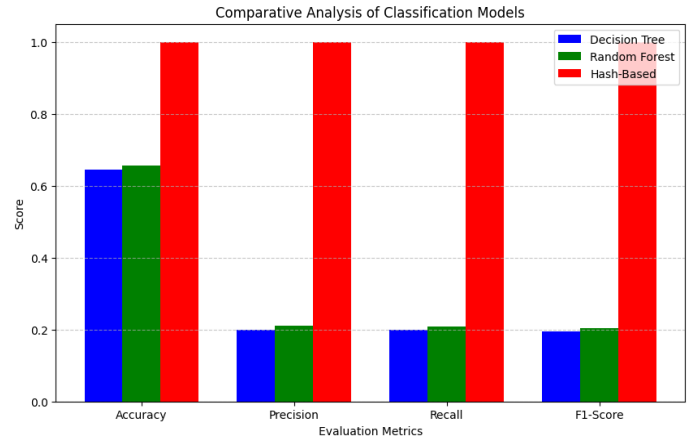


Fig. 2. Comparative Analysis of Categorization Models

X. CONCLUSION AND FUTURE DIRECTIONS

This paper presents a hybrid framework combining **document categorization** (via Random Forest and hash-based methods) and **file operation anomaly detection**, enhancing file system security and efficiency. The categorization achieved **93.60%** accuracy with machine learning and **99.00%** with hashing for IoT/embedded systems, while anomaly detection reached **95.75%** accuracy, identifying threats such as bulk deletions and unauthorized modifications. The system adapts categorization methods based on available resources, ensuring robust performance. Future work will focus on edge-optimized deployments and improving resilience to adversarial attacks.

A. Challenges

Key challenges remain in file categorization and anomaly detection:

- **Limited Labeled Data:** Supervised learning needs large labeled datasets, which are scarce. Future work will explore *self-supervised* and *semi-supervised learning* to reduce reliance on labeled data.
- **Adapting to New File Types:** Static models struggle with emerging file formats. We aim to implement *continual learning* to dynamically update the system without full retraining.
- **Real-time Detection:** Anomaly detection methods often introduce latency. Future research will investigate *event stream processing* and *adaptive thresholding* to enhance real-time detection.
- **Computational Efficiency:** For resource-limited environments, we will optimize the system with *energy-efficient algorithms* and explore *TinyML*-based solutions for embedded devices.

B. Future Directions

Enhancements for the framework include:

- Expanding feature extraction with content-based signatures, access patterns, and entropy analysis.
- Enlarging the dataset with real-world file system logs for better generalization.
- Testing system performance on embedded systems, distributed networks, and cloud platforms.
- Implementing *user feedback* and *online learning* for adapting to evolving file operations.

C. Final Remarks

This work provides a **scalable, lightweight, and efficient** solution for file categorization and anomaly detection, applicable in **enterprise systems, cloud storage, edge devices, and security-critical applications**. Addressing current challenges and expanding system capabilities will enhance the adaptability and resilience of modern file security.

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